

4. Shown below are the number of galleys for a manuscript (X) and the total dollar cost of correcting typographical errors (Y) in a random sample of recent orders handled by a firm specializing in technical manuscripts. Since Y involves variable costs only, an analyst wished to determine whether regression-through-the-origin model (1) is appropriate for studying the relation between the two variables.

i :	1	2	3	4	5	6	7	8	9	10	11	12
X_i :	7	12	10	10	14	25	30	25	18	10	4	6
Y_i :	128	213	191	178	250	446	540	457	324	177	75	107

Note: Model (1) is a straight line through the origin, with slope β_1 .

$$Y_i = \beta_1 X_i + \epsilon_i$$

where:

β_1 is a parameter,

X_i are known constants,

ϵ_i are independent $N(0, \sigma^2)$.

(a) Fit regression model (1) and state the estimated regression function.

(b) Plot the estimated regression function and the data. Does a linear regression function through the origin appear to provide a good fit here? Comment.

(c) In estimating costs of handling prospective orders, management has used a standard of \$17.50 per galley for the cost of correcting typographical errors. Test whether or not this standard should be revised; use $\alpha = .02$. State the alternatives, decision rule, and conclusion.

(d) Obtain a prediction interval for the correction cost on a forthcoming job involving 10 galleys. Use a confidence coefficient of 98 percent.